

Review: The real number system



- 1
- a $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ b $\{0, 1, 2, 3, \dots\}$
c 0
- 2 The set $\{0, 1, 2, \dots\}$ is called the set of whole numbers
- 3
- a
- i False ii False iii True
- b
- i 38 is a rational, an integer and a whole number.
ii -42 is a rational number and an integer.
iii $\sqrt{6}$ is an irrational number
iv 3.058 is a rational number.
v $\sqrt{49}$ is a rational number, an integer, and a whole number.
vi $\frac{\pi}{3}$ is an irrational number.
vii $0.5777\dots$ is a rational number
viii $\frac{26}{45}\dots$ is a rational number
- 4 Every rational number can be expressed in the form $\frac{p}{q}$ where p is an integer and q is a positive integer.
A whole number can be expressed in this form where $q = 1$ and $p \geq 0$. That is to say, a whole number is a non-negative integer.
- 5 The decimal could either be a repeating decimal or a terminating decimal.
- 6
- a Whole numbers b Rational numbers
c Positive real numbers d All real numbers
e Integers f Rational numbers
- 7
- a Yes b No c No d Yes



Additional questions

- 8 The subset of rational numbers includes any number that can be written in the form $\frac{a}{b}$ where a and b are integers and $b \neq 0$, and thus can be written as a terminating or repeating decimal.
- 9 Possible answers:
- The subset of irrational numbers includes any number that cannot be written in the form $\frac{a}{b}$ where a and b are integers, and thus cannot be written as a terminating or repeating decimal.
 - Any negative statement of the answer to question 8 will suffice.
- 10 No. Irrational numbers can be defined by being not rational and so the two subsets of numbers are mutually exclusive.
- 11 The integers belong to the subset of rational numbers, since any integer can be written as a fraction with a denominator of 1.
- 12
- | | | | |
|--------------|--------------|------------|--------------|
| a Rational | b Irrational | c Rational | d Rational |
| e Irrational | f Irrational | g Rational | h Irrational |
- 13
- | | | | |
|---------------|--------------------------------|-----------------------------|------------------|
| a 4, Rational | b $\frac{\pi}{4}$, Irrational | c $-\frac{4}{5}$, Rational | d 0.08, Rational |
|---------------|--------------------------------|-----------------------------|------------------|
- 14 No. Although the decimal 24.737337333... may have a pattern, it cannot be expressed as either a repeating decimal or a terminating decimal, so the number cannot be rational.